

SynOS Frontend Build Playbook - Screen-by-Screen (Days 18-20)

Every Button, Every Field, Every Action Documented

Last Updated: November 22, 2025, 3:48 PM IST

Status: ✓ PRODUCTION READY - BACKEND COMPLETE (Days 1-17)

For: Solo Developer Building Frontend After Complete Backend

Timeline: 3 days (Days 18-20)

Total Screens: 45+ screens across 8 roles

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OVERVIEW: SCREEN-BY-SCREEN STRATEGY

Why Screen-by-Screen?

Your backend is 100% ready. Every API endpoint tested via Postman/Swagger. Zero mocks, zero placeholders.

Frontend strategy:

- Build 1 screen at a time
- Test immediately against real backend
- No component gets skipped
- Every button wired to real API

Build Order (45 Screens in 3 Days)

Day	Screens	Count	Focus
Day 18	Auth + Reception	15 screens	Login, patients, visits, payments
Day 19	Lab Processing	18 screens	Results, signing, reports, delivery
Day 20	Admin + Polish	12 screens	Masters, inventory, radiology, monitoring

DAY 18: CORE SCREENS (AUTHENTICATION + RECEPTION)

Screen 1: Login Screen

Route: /login

Users: All 8 roles

Backend API: POST /api/v1/auth/login

UI Components Needed:

```
SynOS - Login |
|
Email: [] |

```

Complete Button & Field Checklist:

- [] Email input field
 - Type: email
 - Required: Yes
 - Validation: Email format
 - API field: email
- [] Password input field

- Type: password
 - Required: Yes
 - Min length: 8 characters
 - API field: password
- **[] Remember Me checkbox**
 - Optional
 - Stores JWT in localStorage (if checked) vs sessionStorage
- **[] Login button**
 - Text: "Login" or "Sign In"
 - Action: POST /api/v1/auth/login
 - Disabled when: Empty email or password
 - Shows spinner when: API call in progress
- **[] Error display area**
 - Shows: API error messages
 - Examples: "Invalid credentials", "Account locked"
 - Color: Red text
- **[] Loading spinner**
 - Shows: During API call
 - Hides: After success or error

Gemini Prompt for Screen 1:

Build the SynOS Login screen (React + Tailwind + shadcn/ui).

BACKEND API:

- Endpoint: POST /api/v1/auth/login
- Request: { "email": "string", "password": "string" }
- Response: { "accessToken": "string", "refreshToken": "string", "user": { "userId": "uuid", "email": "string", "name": "string", "role": "string" } }
- Error: { "error": { "code": "string", "message": "string" } }

UI REQUIREMENTS:

1. Email input field
 - Type: email
 - Required
 - Validation: Email format
 - Error message: "Please enter a valid email"
2. Password input field
 - Type: password
 - Required
 - Min length: 8
 - Error message: "Password must be at least 8 characters"
3. Remember Me checkbox
 - Optional
 - If checked: Store JWT in localStorage
 - If unchecked: Store JWT in sessionStorage
4. Login button
 - Text: "Login"
 - Disabled: If email or password empty
 - Shows spinner: During API call

- Action: Call POST /api/v1/auth/login
 - On success: Store tokens, redirect to role-specific home
 - On error: Display error message below button
5. Error display area
- Red text
 - Shows API error messages
 - Clear on new input
6. Loading spinner
- Shows during API call
 - Hides after response

REDIRECT LOGIC:

- Reception → /reception/dashboard
- Lab Tech → /lab/worklist
- Pathologist → /pathologist/review-queue
- Sample Collection → /sample/collection
- Delivery → /delivery/queue
- Admin → /admin/dashboard
- Finance → /finance/dashboard
- Radiology → /radiology/worklist

KEYBOARD SHORTCUTS:

- Enter key: Submit login (if form valid)
- Tab: Navigate between fields

DO NOT:

- Use mock data
- Use placeholder APIs
- Skip error handling
- Skip loading states

ACCEPT CRITERIA:

- Login works for all 8 roles
- JWT stored correctly (localStorage or sessionStorage)
- Redirect to correct dashboard per role
- Error messages display for wrong credentials
- Loading spinner shows during API call

Screen 2: Reception Dashboard (Home)

Route: /reception/dashboard

Role: Reception

Backend APIs:

- GET /api/v1/visits?dept=Pathology&status=Registered&limit=10 (today's visits)
- GET /api/v1/appointments?date=2025-11-22 (today's appointments)

UI Components Needed:

SynOS - Reception [Logout]			
Search Patient: [] [Search] (/)			
Quick Actions:			
[New Patient (Ctrl+N)] [New Visit (Ctrl+V)]			
[Appointments (Ctrl+A)]			
Today's Stats:			
Total Visits	Pending Pay	Completed	
42	8	34	
Recent Visits (Last 10):			
Token	Patient Name	Status	Action
P-042	Ramesh Sharma	Paid	[View]
P-041	Priya Singh	Unpaid	[Payment]
...			
Upcoming Appointments (Today):			
Time	Patient Name	Dept	Action
10:30	Anand Kumar	Path	[Check-in]
11:00	Sita Devi	Radio	[Check-in]

Complete Button & Field Checklist:

- [] **Logout button**
 - Text: "Logout"
 - Action: Clear JWT, redirect to /login
 - Position: Top right
- [] **Search patient input**
 - Placeholder: "Search by name, phone, MRN..."
 - Keyboard shortcut: / (focus search)
 - Type: text
 - API: GET /api/v1/patients?search={query}
- [] **Search button**
 - Text: "Search" or magnifying glass icon
 - Action: Navigate to patient search results
- [] **New Patient button**
 - Text: "New Patient"
 - Keyboard: Ctrl+N

- Action: Navigate to /reception/patients/new
- [] **New Visit button**
 - Text: "New Visit"
 - Keyboard: Ctrl+V
 - Action: Navigate to /reception/visits/new
- [] **Appointments button**
 - Text: "Appointments"
 - Keyboard: Ctrl+A
 - Action: Navigate to /reception/appointments
- [] **View button (in Recent Visits table)**
 - Text: "View"
 - Action: Navigate to /reception/visits/{visitId}
 - One per row
- [] **Payment button (in Recent Visits table)**
 - Text: "Payment"
 - Shows only: If status = "Unpaid"
 - Action: Navigate to /reception/payments/{visitId}
- [] **Check-in button (in Appointments table)**
 - Text: "Check-in"
 - Action: Create visit from appointment, navigate to visit screen
 - API: POST /api/v1/visits (with appointmentId)

Gemini Prompt for Screen 2:

Build the Reception Dashboard screen (React + Tailwind + shadcn/ui).

BACKEND APIs:

1. GET /api/v1/visits?dept=Pathology&status=Registered&limit=10
Response: { "data": [{ "visitId": "uuid", "token": "P-042", "patient": { "name": "string", "mrn": "string" }, "status": "Paid|Unpaid", "createdAt": "ISO" }], "pagination": { ... } }
2. GET /api/v1/appointments?date=2025-11-22
Response: { "data": [{ "appointmentId": "uuid", "patient": { "name": "string" }, "scheduledFor": "ISO", "dept": "string" }] }
3. GET /api/v1/patients?search={query}
(For search functionality)

UI REQUIREMENTS:

HEADER:

1. Logo/Title: "SynOS - Reception"
2. Logout button (top right)
 - Action: Clear localStorage/sessionStorage, redirect to /login

SEARCH BAR:

3. Search patient input

- Placeholder: "Search by name, phone, MRN..."
- Keyboard shortcut: / (focus input)
- On Enter: Navigate to /reception/patients/search?q={query}

4. Search button

- Icon: Magnifying glass
- Action: Same as Enter key

QUICK ACTIONS:

5. New Patient button

- Text: "New Patient"
- Keyboard: Ctrl+N
- Navigate to: /reception/patients/new

6. New Visit button

- Text: "New Visit"
- Keyboard: Ctrl+V
- Navigate to: /reception/visits/new

7. Appointments button

- Text: "Appointments"
- Keyboard: Ctrl+A
- Navigate to: /reception/appointments

TODAY'S STATS:

8. Total Visits count (from API)

9. Pending Payment count (filter status=Unpaid)

10. Completed count (filter status=Paid)

RECENT VISITS TABLE:

11. Table with columns: Token, Patient Name, Status, Action

12. View button (each row)

- Navigate to: /reception/visits/{visitId}

13. Payment button (each row, only if status=Unpaid)

- Navigate to: /reception/payments/{visitId}

APPOINTMENTS TABLE:

14. Table with columns: Time, Patient Name, Dept, Action

15. Check-in button (each row)

- Action: POST /api/v1/visits with appointmentId

- On success: Navigate to /reception/visits/{newVisitId}

KEYBOARD SHORTCUTS:

- / : Focus search
- Ctrl+N: New patient
- Ctrl+V: New visit
- Ctrl+A: Appointments
- Ctrl+H: Home (this screen)

DO NOT:

- Use mock data
- Skip any button
- Skip keyboard shortcuts
- Use placeholder APIs

ACCEPT CRITERIA:

- All 15 components present
- All buttons navigate correctly
- API data displayed in tables
- Keyboard shortcuts work
- Loading states for API calls

Screen 3: Patient Search Results

Route: /reception/patients/search?q={query}

Role: Reception

Backend API: GET /api/v1/patients?search={query}&limit=50

UI Components Needed:

SynOS - Patient Search [Home]																												
Search: [Ramesh_____] [Search] (Ctrl+F)																												
Results for "Ramesh": (12 found)																												
<table border="1"><thead><tr><th>MRN</th><th>Name</th><th>DOB</th><th>Phone</th><th>Action</th></tr></thead><tbody><tr><td>A00001</td><td>Ramesh Sharma</td><td>1980-05-15</td><td>9876543</td><td>[Select] </td></tr><tr><td>A00012</td><td>Ramesh S</td><td>1980-05-16</td><td>9876543</td><td>[Select] </td></tr><tr><td>A00045</td><td>Ramesh Kumar</td><td>1982-03-10</td><td>9123456</td><td>[Select] </td></tr><tr><td>...</td><td></td><td></td><td></td><td> </td></tr></tbody></table>				MRN	Name	DOB	Phone	Action	A00001	Ramesh Sharma	1980-05-15	9876543	[Select]	A00012	Ramesh S	1980-05-16	9876543	[Select]	A00045	Ramesh Kumar	1982-03-10	9123456	[Select]	...				
MRN	Name	DOB	Phone	Action																								
A00001	Ramesh Sharma	1980-05-15	9876543	[Select]																								
A00012	Ramesh S	1980-05-16	9876543	[Select]																								
A00045	Ramesh Kumar	1982-03-10	9123456	[Select]																								
...																												
[Create New Patient (Ctrl+N)]																												
Possible Duplicates Detected:																												
△ A00001 and A00012 may be same person (95% match)																												
[Merge Patients]																												

Complete Button & Field Checklist:

- [] **Home button**
 - Text: "Home"
 - Action: Navigate to /reception/dashboard
- [] **Search input**
 - Prefilled with query param
 - Keyboard: Ctrl+F (focus)
 - Type: text
- [] **Search button**
 - Text: "Search"
 - Action: Reload with new query
- [] **Select button (each row)**
 - Text: "Select"
 - Action: Navigate to /reception/patients/{patientId}

- One per patient
- [] **Create New Patient button**
 - Text: "Create New Patient"
 - Keyboard: Ctrl+N
 - Action: Navigate to /reception/patients/new
- [] **Merge Patients button**
 - Shows only: If duplicates detected
 - Text: "Merge Patients"
 - Action: Navigate to /reception/patients/merge?source={id1}&target={id2}

Gemini Prompt for Screen 3:

Build the Patient Search Results screen (React + Tailwind + shadcn/ui).

BACKEND API:

- GET /api/v1/patients?search={query}&limit=50
Response: { "data": [{ "patientId": "uuid", "mrn": "A00001", "name": "Ramesh Sharma", "dob": "1980-05-15", "phone": "9876543210" }], "pagination": { ... } }
- GET /api/v1/patients/{id}/possible-duplicates
Response: { "data": [{ "patientId": "uuid", "mrn": "string", "matchScore": 95, "matchReason": "string" }] }

UI REQUIREMENTS:

HEADER:

1. Home button (top right)
 - Navigate to: /reception/dashboard

SEARCH:

2. Search input

- Prefilled with URL query param "q"
- Keyboard: Ctrl+F (focus)

3. Search button
 - Action: Reload page with new query

RESULTS TABLE:

4. Table columns: MRN, Name, DOB, Phone, Action

5. Result count: "X found"

6. Select button (each row)
 - Navigate to: /reception/patients/{patientId}

NEW PATIENT:

7. Create New Patient button

- Keyboard: Ctrl+N
- Navigate to: /reception/patients/new

DUPLICATE DETECTION:

8. Call GET /api/v1/patients/{firstResultId}/possible-duplicates

9. If duplicates found (matchScore > 85):

- Show warning: "⚠ A00001 and A00012 may be same person (95% match)"

10. Merge Patients button (only if duplicates)

- Navigate to: /reception/patients/merge?source={id1}&target={id2}

KEYBOARD SHORTCUTS:

- Ctrl+F: Focus search
- Ctrl+N: New patient
- Esc: Go back to dashboard

DO NOT:

- Skip duplicate detection
- Use mock data
- Skip merge button when duplicates exist

ACCEPT CRITERIA:

- Search returns real results
- Duplicate detection works
- All buttons functional
- Keyboard shortcuts work

Screen 4: New Patient Registration

Route: /reception/patients/new

Role: Reception

Backend API: POST /api/v1/patients

UI Components Needed:

SynOS - New Patient Registration [Cancel]
Patient Information:
Full Name: [] *
<i>Date of Birth: [DD] / [MM] / [YYYY] *</i>
<i>Age: [Auto-calculated from DOB]</i>
<i>Sex: () Male () Female () Other *</i>
<i>Phone: [] *</i>
<i>Address: []</i>

City:
State:
PIN Code:
Email:
[Save Patient (Ctrl+S)] [Cancel]
Error/Success message area

Complete Button & Field Checklist:

- ☐ **Cancel button**
 - Text: "Cancel"
 - Action: Navigate back or to /reception/dashboard
- ☐ **Full Name input**
 - Required: Yes (*)
 - Type: text
 - Validation: Min 2 characters
 - API field: name
- ☐ **Date of Birth inputs**
 - Required: Yes (*)
 - Three dropdowns: DD, MM, YYYY
 - Validation: Valid date, not future date
 - API field: dob (format: YYYY-MM-DD)
- ☐ **Age display**
 - Auto-calculated from DOB
 - Read-only
 - API field: age (calculated)
- ☐ **Sex radio buttons**
 - Required: Yes (*)
 - Options: Male, Female, Other
 - API field: sex (M/F/O)
- ☐ **Phone input**
 - Required: Yes (*)
 - Type: tel
 - Validation: 10 digits
 - API field: phone
- ☐ **Address input**
 - Optional
 - Type: text
 - API field: address
- ☐ **City input**
 - Optional
 - Type: text
 - API field: city
- ☐ **State input**
 - Optional

- Type: text or dropdown (Indian states)
 - API field: state
- **[] PIN Code input**
 - Optional
 - Type: text
 - Validation: 6 digits
 - API field: pinCode
- **[] Email input**
 - Optional
 - Type: email
 - Validation: Email format
 - API field: email
- **[] Save Patient button**
 - Text: "Save Patient"
 - Keyboard: Ctrl+S
 - Action: POST /api/v1/patients
 - Disabled when: Required fields empty or invalid
 - Shows spinner during API call
- **[] Error/Success message area**
 - Shows: API errors or success message
 - On success: "Patient created! MRN: A00056"
 - On error: Display error details

Gemini Prompt for Screen 4:

Build the New Patient Registration screen (React + Tailwind + shadcn/ui).

BACKEND API:

- POST /api/v1/patients
 Request: {
 "name": "string",
 "dob": "YYYY-MM-DD",
 "sex": "M|F|O",
 "phone": "string",
 "address": "string",
 "city": "string",
 "state": "string",
 "pinCode": "string",
 "email": "string"
 }
 Response: { "data": { "patientId": "uuid", "mrn": "A00056", "name": "string", ... } }
 Error: { "error": { "code": "string", "message": "string", "details": [...] } }

UI REQUIREMENTS:

HEADER:

1. Cancel button (top right)
 - Navigate back to previous screen or /reception/dashboard

FORM FIELDS:

2. Full Name input

- Required (*)
- Validation: Min 2 characters
- Error: "Name must be at least 2 characters"

3. Date of Birth (3 dropdowns: DD, MM, YYYY)

- Required (*)
- Validation: Valid date, not future
- Error: "Invalid date" or "DOB cannot be in future"

4. Age display (read-only, auto-calculated from DOB)

5. Sex radio buttons

- Required (*)
- Options: Male, Female, Other
- Error: "Please select sex"

6. Phone input

- Required (*)
- Validation: Exactly 10 digits
- Error: "Phone must be 10 digits"

7. Address input (optional)

8. City input (optional)

9. State input (optional, dropdown preferred with Indian states)

10. PIN Code input (optional, 6 digits validation)

11. Email input (optional, email format validation)

ACTIONS:

12. Save Patient button

- Keyboard: Ctrl+S
- Disabled: If required fields invalid
- Shows spinner: During API call
- On success: Show "Patient created! MRN: {mrn}", navigate to /reception/patients/{patientId}
- On error: Display error message

13. Cancel button

- Navigate back

VALIDATION:

- Real-time validation on blur
- Show error messages below each field
- Disable Save button if any required field invalid

KEYBOARD SHORTCUTS:

- Ctrl+S: Save
- Esc: Cancel

DO NOT:

- Skip required field validation
- Allow future DOB

- Skip phone number validation
- Use mock API

ACCEPT CRITERIA:

- All 13 components present
- All validations work
- API call creates patient
- MRN displayed on success
- Navigation works

Screens 5-15: Remaining Day 18 Screens

I'll provide complete checklists for all remaining Day 18 screens:

5. **Patient Details** (/reception/patients/{id})
6. **Patient Merge** (/reception/patients/merge)
7. **New Visit** (/reception/visits/new)
8. **Visit Details** (/reception/visits/{id})
9. **Payment** (/reception/payments/{visitId})
10. **Invoice/Receipt Print** (modal/print view)
11. **Appointments List** (/reception/appointments)
12. **New Appointment** (/reception/appointments/new)
13. **Sample Collection** (/sample/collection)
14. **Barcode Print** (modal/print view)
15. **Token Print** (modal/print view)

Would you like me to continue with the complete detailed breakdown for each screen (5-15), then move to Day 19 and Day 20 screens?

Each screen will have:

- Complete UI layout
- Every button/field/component listed
- API endpoints
- Validation rules
- Keyboard shortcuts
- Complete Gemini prompt

Confirm if you want all 45 screens documented this way, or would you prefer a different format?